

Monetary Economics and Policy

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Chapter 4: Private Money



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What We Will Learn

- ▶ How private intermediaries create **safe, money-like assets** and how these coexist with government-issued liquidity.
- ▶ Why private money creation does *not* undermine the central bank's **control of the price level**, thanks to interest on reserves and the backing principle.
- ▶ How competition eliminates liquidity premia, driving the economy toward **liquidity satiation** and zero rents.
- ▶ Under which conditions different institutional regimes (free banking, narrow banking, real-bills-type arrangements) yield **equivalent efficient allocations**.
- ▶ Why, in a multi-currency environment, the currency with **stable prices** becomes the dominant unit of account for safe assets, and how this affects welfare.

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- 4 Public vs. Private Liquidity Supply
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Introduction

- ▶ **Money vs. Currency:** Money is a claim on currency that shares similar *properties* (store of value, medium of exchange). Currency is money, but money is not necessarily currency.
- ▶ Government liabilities (cash, reserves, default-free bonds) are natural providers of liquidity, but **private intermediaries** can also create *money claims* such as deposits.
- ▶ Classic insight from Brunner (1989): financial intermediation creates a wedge between the *monetary base* and the *money stock*, because private claims also provide liquidity services.

Introduction

- ▶ By paying *interest on reserves*, the central bank can separate the achievement of the inflation target from the efficient supply of liquidity.
- ▶ **Backing principle:** price-level determination requires a real anchor, either through
 - ▶ high-quality assets on the central bank's balance sheet, or
 - ▶ fiscal backing (taxation) supported by treasury–central bank cooperation.
- ▶ The same logic applies to **liquidity supply**: abundant liquidity requires sufficient backing.
- ▶ Insufficient backing (few high-quality assets or weak fiscal capacity) can generate inefficiencies and instability.
- ▶ The government, if a monopoly supplier of liquidity, may maintain rents, giving the private sector an incentive to create money claims, *safe assets*, to arbitrage these rents.

Outline of the Results

- ▶ **Intermediaries can engineer safe assets** by backing risky investments with sufficient equity, ensuring safety even in the worst state.
- ▶ **Free entry leads to zero rents:** competition expands private safe-asset supply until the liquidity premium is driven to zero (satiation).
- ▶ **Price-level control is preserved:** the central bank can apply the principles of Chapter 1; private liquidity creation does *not* impair determinacy.
- ▶ **Equivalence:** under ideal conditions, narrow banking, free banking, and “real bills” arrangements can all implement the efficient allocation.
- ▶ **Dominant currency:** when safe assets can be denominated in multiple currencies, the currency with *stable prices* becomes the dominant unit of account.

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Model Uncertainty

- ▶ Let s_t a state at date t with $s_t \in \mathcal{S}$; s^t is a history such that $s^t = (s_t, s_{t-1}, \dots, s_{t_0})$.
- ▶ The conditional density of s^t has the Markov property:
 $f(s^t|s_{t_0}) = f(s_t|s_{t-1})f(s_{t-1}|s_{t-2}) \cdots f(s_{t_0})$.
- ▶ We will use the notation $X_t \equiv X(s^t)$ for a generic stochastic process X .
- ▶ Conditional expectation at t_0 :

$$\mathbb{E}_{t_0} X_t = \sum_{s^t} f(s^t|s_{t_0}) X(s^t).$$

Capital and Technology

- ▶ A fixed capital K is available to the consumers and intermediaries.
- ▶ Output $Y_t = A_t K$, with $A_t \in [A_{\min}, A_{\max}]$.
- ▶ Denote with Q_t^K the price of capital (in units of currency), the nominal gross return of capital at time t is equal to:

$$1 + i_t^K \equiv \frac{Q_t^K + P_t A_t}{Q_{t-1}^K}.$$

Consumers: Utility

- ▶ The representative agent maximizes

$$\mathbb{E}_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left\{ C_t + V\left(\frac{B_t + S_t}{P_t}\right) \right\}, \quad 0 < \beta < 1. \quad (4.1)$$

- ▶ Utility is linear in consumption C_t ; P_t is the price of the consumption good.
- ▶ The term $V(\cdot)$ captures the utility benefit from holding **safe assets**. It is concave and features a *satiation point* $\bar{q} > 0$ such that

$$V_q(q_t) = 0 \quad \text{for } q_t \geq \bar{q}, \quad q_t \equiv \frac{B_t + S_t}{P_t}.$$

- ▶ Safe assets are nominal, risk-free claims that provide **liquidity services** (e.g. transaction value or collateral value). They include:
 - ▶ B_t : government liabilities (including reserves),
 - ▶ S_t : risk-free securities issued by intermediaries.

Consumer: Budget Constraint

The representative consumer faces the following **budget constraint**:

$$B_t + S_t + Q_t^K K_t + N_t + P_t C_t + T_t = (1 + i_{t-1}^X)B_{t-1} + (1 + i_{t-1}^S)S_{t-1} + (1 + i_t^K)Q_{t-1}^K K_{t-1} + \Psi_t^D. \quad (4.2)$$

- ▶ The risky asset K_t has price Q_t^K and a stochastic nominal return $1 + i_{t+1}^K$; the safe assets earn $1 + i_t^X$ and $1 + i_t^S$.
- ▶ Consumers finance intermediaries' equity N_t and receive dividends Ψ_{t+1}^D .
- ▶ The government levies lump-sum taxes T_t .
- ▶ The consumer chooses sequences

$$\{C_t, B_t, S_t, K_t, N_t\}_{t=t_0}^{\infty}, \quad C_t, B_t, S_t, N_t \geq 0,$$

to maximize (4.1) subject to (4.2) and a borrowing limit at each $t \geq t_0$, given initial conditions:

$$(1 + i_{t_0-1}^X)B_{t_0-1}, \quad (1 + i_{t_0-1}^S)S_{t_0-1}, \quad (1 + i_{t_0-1}^K)K_{t_0-1}.$$

Consumer: First-Order Conditions

The FOC for consumption implies $\lambda_t = 1/P_t$, where λ_t is the multiplier on (4.2). Define the stochastic discount factor $\tilde{R}_{t,t+1} \equiv \beta P_t/P_{t+1}$. The respective first-order conditions follow:

► **Capital:**

$$1 = \mathbb{E}_t \{ \tilde{R}_{t,t+1} (1 + i_{t+1}^K) \} . \quad (4.3)$$

► **Equity valuation:**

$$N_t = \mathbb{E}_t \{ \tilde{R}_{t,t+1} \Psi_{t+1}^D \} . \quad (4.4)$$

► **Private safe debt:**

$$1 \geq \varrho_t + \mathbb{E}_t \{ \tilde{R}_{t,t+1} (1 + i_t^S) \} . \quad (4.5)$$

► **Government safe debt:**

$$1 \geq \varrho_t + \mathbb{E}_t \{ \tilde{R}_{t,t+1} (1 + i_t^X) \} . \quad (4.6)$$

► **Liquidity premium:** $\varrho_t \equiv V_q \left(\frac{B_t + S_t}{P_t} \right)$.

Consumers: First-Order Conditions

Notes:

- ▶ Equations (4.5) and (4.6) hold with equality when the corresponding asset is used for liquidity purposes. In this case, its price (one dollar) equals the discounted value of its payoff *plus* the non-pecuniary liquidity services.
- ▶ At least one of (4.5)–(4.6) must bind, otherwise safe assets would not provide liquidity services in equilibrium.
- ▶ The consumer's problem is closed by the intertemporal budget constraint.

Consolidated Budget Constraint

- ▶ We assume that the treasury is fully backed by the central bank; therefore, we do not distinguish between the two agents.
- ▶ The **consolidated government budget constraint** is:

$$B_t = (1 + i_{t-1}^X)B_{t-1} - T_t. \quad (4.7)$$

- ▶ Because all government liabilities are backed by the central bank, B_t is effectively risk free and not subject to a solvency condition.

Intermediaries

- ▶ **Intermediaries** live for two periods.
- ▶ At time t they invest in risky capital K^I with price Q^K and finance it by issuing safe debt S and equity N .
- ▶ They are therefore subject to the following **budget constraint**:

$$Q_t^K K_t^I = S_t + N_t \quad (4.8)$$

- ▶ At time $t + 1$ **gross profits** are

$$\Psi_{t+1}^I = (1 + i_{t+1}^K)Q_t^K K_t^I - (1 + i_t^S)S_t \quad (4.9)$$

Limited Liability

- ▶ Profits cannot be negative due to **limited liabilities**, therefore:

$$(1 + i_{min}^K)Q_t^K K_t^I \geq (1 + i_t^S)S_t, \quad (4.10)$$

in which i_{min}^K is the worst realization of i_t^K .

- ▶ Substituting (4.8) into (4.10) for $Q_t^K K_t^I$ implies an equity lower bound

$$N_t \geq \bar{N}_t = \left(\frac{1 + i_t^S}{1 + i_{min}^K} - 1 \right) S_t. \quad (4.11)$$

- ▶ Assets are risky; intermediaries should back safe debt with an appropriate level of equity to absorb losses in the worst-case scenario.

Zero-Rent Supply Schedule

- Intermediaries maximize expected rents $\mathcal{R}$:

$$\mathcal{R}_t = E_t \left\{ \tilde{R}_{t,t+1} (\Psi_{t+1}^I - \Psi_{t+1}^D) \right\} = S_t - E_t \left\{ \tilde{R}_{t,t+1} (1 + i_t^S) \right\} S_t,$$

having used (4.3), (4.4), (4.8) and (4.9) to obtain the second equality.

- Maximization implies that intermediaries supply a positive amount of securities when rents are positive, i.e.,

$$\frac{1}{1 + i_t^S} \geq E_t \left\{ \tilde{R}_{t,t+1} \right\}.$$

Zero-Rent Supply Schedule

- ▶ Because entry into the securities market is free and an infinite number of intermediaries can potentially enter, rents are zero in equilibrium. This implies the pricing condition:

$$\frac{1}{1 + i_t^S} = \mathbb{E}_t \{ \tilde{R}_{t,t+1} \} . \quad (4.12)$$

- ▶ Intermediaries creating “safe assets” choose equity to satisfy the lower bound in (4.11) and supply securities at the price given by (4.12).

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Market Clearing and Liquidity Satiation

Goods market clearing:

$$C_t = Y_t = A_t K$$

Capital market clearing:

$$K_t + K_t^I = K$$

Bonds market:

- ▶ Bonds issued by the government B_t and safe assets issued by intermediaries S_t are held by households.

Key result:

- ▶ Combining demand (4.6) with supply (4.12) implies $\varrho_t = 0$ in equilibrium.
- ▶ If $\varrho_t > 0$, entry is profitable; competition expands S_t until the premium is eliminated.
- ▶ $\varrho_t = 0$: the economy is at **liquidity satiation**.

Intertemporal Resource Constraints

- Combining the household intertemporal budget constraint with the balance sheets of intermediaries yields the intertemporal **resource** constraint:

$$\frac{(1 + i_{t_0-1}^X)B_{t_0-1}}{P_{t_0}} = \mathbb{E}_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t}{P_t} + \frac{i_t - i_t^X}{1 + i_t} \frac{B_t}{P_t} \right) \right\}. \quad (4.13)$$

- i_t is the return on a notional risk-free bond that does not provide liquidity and is priced as:

$$\frac{1}{1 + i_t} = \mathbb{E}_t \{ \tilde{R}_{t,t+1} \}. \quad (4.14)$$

- Real value of debt should be equal to the expected present-discounted value of taxes plus liquidity rents obtained by government- issued “safe assets”.

Government Policy

Assume the following **government policy**:

- ▶ The interest rate on reserves is constant and satisfies:

$$1 + i_t^X = \beta^{-1}.$$

- ▶ Tax policy is defined in real terms as:

$$\frac{T_t}{P_t} = (1 - \beta)\tau, \quad \tau > 0.$$

- ▶ Since in equilibrium $q_t = 0$, it must be that

$$1 + i_t = 1 + i_t^X = \beta^{-1}.$$

Price Level Determination

- Using tax policy and $1 + i_t = 1 + i_t^X = \beta^{-1}$ in (4.13) determines the price level P_{t_0} given B_{t_0-1} , i_{t_0-1} , and τ :

$$\frac{(1 + i_{t_0-1})B_{t_0-1}}{P_{t_0}} = \tau \quad \Rightarrow \quad P_{t_0} = \frac{(1 + i_{t_0-1})B_{t_0-1}}{\tau}.$$

- Repeating the same substitution for future contingencies implies that $\{P_t\}_{t_0}^\infty$ is a deterministic sequence.

Price Level Determination

- From (4.14):

$$\frac{1}{1+i_t} = \mathbb{E}_t\{\tilde{R}_{t,t+1}\} = \beta \frac{P_t}{P_{t+1}}.$$

- Since $1+i_t = 1+i_t^X = \beta^{-1}$, this condition implies

$$P_{t+1} = P_t,$$

so the price level remains constant at some level P .

- **Key result:** Private creation of safe assets does not undermine the central bank's control of the price level.

Equilibrium Liquidity Supply

- ▶ In equilibrium $\varrho_t = 0$, so the economy is at full satiation of liquidity.
- ▶ Government liquidity is fixed each period at:

$$\frac{B_t}{P_t} = \beta\tau.$$

- ▶ The private supply of liquidity adjusts to ensure overall satiation:

$$S \geq \max\{P(\bar{q} - \beta\tau), 0\}.$$

- ▶ If government liquidity is insufficient to reach satiation, i.e. $\beta\tau < \bar{q}$, intermediaries have an incentive to issue additional safe assets so that:

$$S \geq P(\bar{q} - \beta\tau).$$

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Private Liquidity Supply

Hayek (1976):

- ▶ Competition leads the private sector to supply a sufficiently large quantity of liquidity (safe assets).
- ▶ A positive liquidity premium induces entry; free competition expands S until $\varrho = 0$ (full satiation), supported by sufficient equity backing.
- ▶ Unfettered competition achieves efficiency without requiring regulation.
- ▶ **Real Bills Doctrine:** Intermediaries should hold safe assets to back private money. However, this is **not necessary**. Competition for supplying safe securities already forces intermediaries to raise enough equity to absorb potential losses.

Narrow Banking

Friedman (1960):

- ▶ Argued that the government should monopolize the supply of liquidity.
- ▶ Intermediaries would face a 100% reserve requirement; liquid asset supply would be determined solely by government debt B .
- ▶ With sufficient taxes T to back interest payments, this policy can still achieve $\beta\tau \geq \bar{q}$.
- ▶ Alternatively, public liquidity can be supplied by the central bank through asset investment.
- ▶ If the central bank transforms risk-free but illiquid private assets into liquid ones, this can be achieved without raising taxes.
- ▶ If it invests only in risky private securities, losses in default states must be covered by taxation, creating monetary–fiscal linkages.

Bottom Line

- ▶ Both *competition* in banking sector and *centralized public liquidity supply* can deliver efficiency without jeopardizing price-level control.
- ▶ As emphasized by Hayek (1976), a monopoly “prevents the discovery of better methods of satisfying a need for which a monopolist has no incentive”. It could be that the government may not have incentives to satiate liquidity.
- ▶ If equity can be raised without frictions, competition by private suppliers of liquidity can achieve the efficient allocation.

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Dominant Currency

- ▶ Chapter 2 analyzed currency competition in terms of *medium-of-exchange* use.
- ▶ Currencies can also compete as a *store of value* and as a *unit of account*.
- ▶ The next model studies competition between currencies in their role as *safe assets*.

Setup

Two agents:

- ▶ Consumers invest in safe assets, which provide liquidity services in the next period.
- ▶ Intermediaries issue safe assets by investing in risky real projects.

Two currencies:

- ▶ Government currency.
- ▶ Private currency.

Safe assets:

- ▶ Safe assets can be issued in either one of the two currencies.

Consumer: Utility

Household utility for consumption and real safe-asset balances:

$$\mathbb{E}_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left\{ C_t + V \left(\frac{S_t}{P_t} + \frac{S_t^*}{P_t^*} \right) \right\}, \quad (4.15)$$

with

- ▶ consumption good C ,
- ▶ price of consumption good in government currency P ,
- ▶ price of consumption good in private currency P^* ,
- ▶ safe asset in government currency S and
- ▶ safe asset in private currency S^* .

Consumer: Budget Constraint

Consumption C is subject to the following budget constraint (in units of goods):

$$\frac{S_t}{P_t} + \frac{S_t^*}{P_t^*} + C_t = \frac{(1 + i_{t-1}^S)S_{t-1}}{P_t} + \frac{(1 + i_{t-1}^{S^*})S_{t-1}^*}{P_t^*} + Y + \Psi_t^A, \quad (4.16)$$

where

- ▶ i^S and i^{S^*} are the returns of the two securities in the respective currencies,
- ▶ Y is a constant endowment and
- ▶ Ψ^A are state-contingent intermediaries' aggregate real profits.
- ▶ Consumption and portfolio choices follow from the maximization of (4.15) under constraint (4.16) and an appropriate borrowing-limit condition.

Safe Asset Demand and Liquidity Value

- ▶ The consumer's first-order conditions imply the demand for safe assets (government and private):

$$1 \geq \varrho_t + \mathbb{E}_t \{ \tilde{R}_{t,t+1} (1 + i_t^S) \}, \quad (4.17)$$

$$1 \geq \varrho_t + \mathbb{E}_t \{ \tilde{R}_{t,t+1}^* (1 + i_t^{S*}) \}. \quad (4.18)$$

- ▶ The liquidity premium is

$$\varrho_t = V_q \left(\frac{S_t}{P_t} + \frac{S_t^*}{P_t^*} \right),$$

and the stochastic discount factors are

$$\tilde{R}_{t,t+1} = \beta \frac{P_t}{P_{t+1}}, \quad \tilde{R}_{t,t+1}^* = \beta \frac{P_t^*}{P_{t+1}^*}.$$

- ▶ Both securities share the same liquidity premium, but their prices may differ due to different inflation paths in the two currencies.

Intermediaries

- ▶ Intermediaries live for two periods and invest in a risky project.
- ▶ Each project costs one unit at time t and yields payoff $\frac{A_{t+1}}{\beta}$ at $t + 1$, with $A_{t+1} \in [A_{\min}, A_{\max}]$ and $\mathbb{E}_t(A_{t+1}) = 1$.
- ▶ Investment is financed by issuing nominal risk-free securities D_t and D_t^* .
- ▶ Intermediaries cannot raise equity to cover losses and rely on liquidity premia to reduce funding costs.
- ▶ The intermediary's budget constraint is:

$$1 = \frac{D_t}{P_t} + \frac{D_t^*}{P_t^*}, \quad (4.19)$$

and profits at time $t + 1$ are:

$$\Psi_{t+1}^I = \frac{A_{t+1}}{\beta} - \frac{(1 + i_t^S)D_t}{P_{t+1}} - \frac{(1 + i_t^{S*})D_t^*}{P_{t+1}^*}.$$

Safety Constraint

- ▶ To issue safe securities, intermediaries must always be able to meet their obligations, i.e., $\Psi_{t+1}^I \geq 0$.
- ▶ Without equity, this requirement imposes an *upper bound* on the interest rate they can offer on safe debt.
- ▶ The bound reflects the compensation for the worst realizations of technology and price levels.
- ▶ The safety condition is:

$$\min \Psi_{t+1}^I = \frac{A_{\min}}{\beta} - \frac{(1 + i_t^S)D_t}{P_{\min}} - \frac{(1 + i_t^{S*})D_t^*}{P_{\min}^*} \geq 0, \quad (4.20)$$

given that prices and technology have the following supports:

$$P_{t+1} \in [P_{\min}, P_{\max}], \quad P_{t+1}^* \in [P_{\min}^*, P_{\max}^*], \quad A_{t+1} \in [A_{\min}, A_{\max}].$$

Intermediaries' Optimization

- Intermediaries maximize expected discounted real profits:

$$E_t[\beta\Psi_{t+1}^I] = \left[1 - E_t\left(\frac{\beta(1+i_t^S)}{P_{t+1}}\right) D_t - E_t\left(\frac{\beta(1+i_t^{S*})}{P_{t+1}^*}\right) D_t^* \right],$$

choosing non-negative D_t, D_t^* , subject to constraints (4.19) and (4.20) and taking market prices as given.

- Competition makes the safety constraints (4.20) binding and the interest rates on the safe securities equal to the maximum rate consistent with safety, defined on the next slide.

Equilibrium Interest Rates

- Combining (4.19) with the safety constraint (4.20), and setting either $D_t = 0$ or $D_t^* = 0$ to isolate each market, yields the **maximum interest rates** that keep intermediaries' debt safe:

$$\frac{1}{1 + \tilde{i}_t^S} = \frac{\beta}{A_{\min}} \frac{P_t}{P_{\min}}, \quad (4.21)$$

$$\frac{1}{1 + \tilde{i}_t^{S^*}} = \frac{\beta}{A_{\min}} \frac{P_t^*}{P_{\min}^*}. \quad (4.22)$$

- These are the reservation prices—the interest rate bounds needed to ensure safety:
 - A lower $A_{\min}$ reduces the minimum project payoff, requiring a *lower* nominal interest rate to keep debt risk-free.
 - Lower $P_{\min}$ or $P_{\min}^*$ increases the real value of repayment in bad states, again requiring *lower* nominal rates to maintain safety.

Example

- ▶ Prices in the government currency are stable at $P_{t+1} = P$, while private-currency prices vary on $[P_{\min}^*, P_{\max}^*]$.
- ▶ We conjecture that in equilibrium intermediaries issue only in the stable-price (government) currency.
- ▶ Competition pushes the government-currency interest rate to its maximum safe value,

$$1 + i_t^S = 1 + \tilde{i}_t^S,$$

which, combined with (4.17) and the constant P_t , implies a liquidity premium

$$\varrho_t = 1 - A_{\min} > 0.$$

- ▶ Since $\varrho_t = V_q\left(\frac{S_t}{P}\right)$, the supply of government-denominated safe assets is constant and given by

$$S = P V_q^{-1}(1 - A_{\min}).$$

Equilibrium

- ▶ Liquidity is not fully satiated because securities require a premium to compensate for the risk of the underlying assets.
- ▶ In equilibrium, safe-asset supply satisfies

$$S_t = n_t D_t, \quad S_t^* = n_t^* D_t^*,$$

where n_t and n_t^* denote the mass of intermediaries issuing in each currency.

- ▶ Using $D_t^* = 0$ in (4.19) gives $D_t = P$, and therefore

$$n_t = \frac{S_t}{D_t} = V_q^{-1}(1 - A_{\min}).$$

Lack of Private Currency Supply

- ▶ Private-currency safe assets are not supplied because the **market interest rate exceeds the maximum rate consistent with safety**:

$$1 + i_t^{S^*} > 1 + \tilde{i}_t^{S^*}.$$

- ▶ At the equilibrium liquidity premium $\varrho_t = 1 - A_{\min}$, the demand condition implies:

$$\frac{1}{1 + i_t^{S^*}} = \frac{\beta}{A_{\min}} \mathbb{E}_t \left\{ \frac{P_t^*}{P_{t+1}^*} \right\}.$$

- ▶ Comparing this with the safe threshold (4.22) and using

$$\frac{1}{P_{\min}^*} > \mathbb{E}_t \left\{ \frac{1}{P_{t+1}^*} \right\},$$

we obtain:

$$1 + i_t^{S^*} > 1 + \tilde{i}_t^{S^*}.$$

- ▶ **Conclusion:** The private currency cannot support safe assets because required interest rates violate the safety constraint.

Price Stability and Safe Assets

- ▶ Price stability allows a currency to dominate as the *unit of account* for safe assets.
- ▶ With non-risky investment technology ($A_{\min} = A_{\max} = 1$), stable prices eliminate balance-sheet mismatches for intermediaries, so no liquidity premium is required to guarantee safety.
- ▶ Competition among investors then drives the liquidity premium to zero, lowering the demand price of safe securities and crowding out issuance in alternative currencies.

Welfare Implications

- ▶ Welfare declines as the liquidity premium ϱ increases, since a higher premium implies a lower supply of safe assets for a given demand.
- ▶ Price stability minimizes the premium required for safety; safe-asset supply (and thus welfare) is maximized in this environment.
- ▶ This complements Chapter 2: low inflation fosters currency dominance in the medium-of-exchange role, while here **stable** prices foster dominance when safe assets serve as means of payment and units of account.

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Main Takeaways

- ▶ Private intermediaries can create safe, **money-like assets** by backing risky investments with sufficient equity or liquidity premia.
- ▶ Free entry drives the liquidity premium to zero, achieving **liquidity satiation** and eliminating rents.
- ▶ Price-level control remains fully with the government/central bank; **private liquidity creation does not undermine determinacy**.
- ▶ **Efficient allocations** can be implemented under both public and private liquidity provision (narrow banking vs. free banking).
- ▶ In a multi-currency environment, the currency with **stable prices** becomes the **dominant unit of account** for safe assets.
- ▶ **Price stability** minimizes the liquidity premium, expands safe-asset supply, and improves welfare.

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